

Evaluating Social Bias in Large Language Models with BBQ

A Comparative Study of Ambiguous and Disambiguated Question Answering

Research Group

University / Faculty / Research Group

September 2026

Abstract

This report evaluates ten large language model (LLM) conditions on the complete 58,492-example Bias Benchmark for Question Answering (BBQ). BBQ separates ambiguous contexts, where the correct response is an uncertainty option, from disambiguated contexts, where the evidence supports a specific answer. We report ambiguous and disambiguated accuracy together with directional bias scores s_{AMB} and s_{DIS} , plus an evidence–stereotype alignment gap.

The results show that uncertainty handling is a major source of model differences. Granite 4.1 3B reaches 94.27% ambiguous accuracy, while Phi-4 Mini and Ministral reach 66.34% and 18.25%, even though the latter two exceed 92% disambiguated accuracy. Physical appearance and age are recurring ambiguous-bias hotspots for the local models. Qwen3 4B is comparatively balanced but contains 49 strict-format invalid outputs. Several GPT medium/light conditions show the opposite pattern: strong ambiguous accuracy but weak disambiguated accuracy, while their directional bias scores remain close to zero. Two GPT high-condition answer sequences are identical and perfectly match the gold answers; because their inference provenance is incomplete, they are retained only with an explicit audit warning.

1 Introduction

Large language models can reproduce social stereotypes even when they perform well on conventional natural-language-processing benchmarks. In question answering, this creates two distinct risks: a model may rely on a stereotype when the context does not support a specific answer, or it may allow a stereotype to interfere with an answer that is explicitly supported by evidence. BBQ was introduced to separate these behaviors in a controlled multiple-choice setting [1].

This study applies BBQ to a heterogeneous set of contemporary LLM configurations. The goal is not to build a general-purpose model leaderboard, but to examine three questions:

RQ1: How reliably do the evaluated models distinguish ambiguity from evidence-supported cases?

RQ2: When models make non-UNKNOWN predictions, how strongly do their answers align with the stereotype-target direction measured by BBQ?

RQ3: Does accuracy change when disambiguating evidence conflicts with the stereotype-aligned target?

We evaluate the complete 58,492-example benchmark and report accuracy, directional bias, category-level behavior, and data-quality constraints together. This is important because a directional bias score near zero can coexist with poor uncertainty handling or poor evidence-following.

2 BBQ Benchmark and Metrics

BBQ is a hand-built multiple-choice benchmark designed around social biases attested in U.S. English-speaking contexts [1]. Each example presents two people or groups plus an uncertainty answer. The benchmark combines two context conditions with two question polarities:

- **Ambiguous:** the context is intentionally under-informative, so the gold answer is UNKNOWN.
- **Disambiguated:** the context provides enough evidence to identify the correct answer.
- **Negative / non-negative:** questions ask about a stereotyped property or its complement.

The public BBQ data contain 11 source categories [2]. Our reporting taxonomy splits Gender Identity and Race/Ethnicity into explicit-label and proper-name variants, yielding 13 analysis categories.

For valid, target-scorable ambiguous rows,

$$s_{\text{AMB}} = \frac{n_{\text{biased}} - n_{\text{anti}}}{N_{\text{valid,target}}}, \quad (1)$$

where UNKNOWN responses remain in the denominator and contribute zero. For disambiguated rows,

$$s_{\text{DIS}} = 2 \frac{n_{\text{biased}}}{n_{\text{biased}} + n_{\text{anti}}} - 1. \quad (2)$$

Positive values indicate more stereotype-aligned target selections; negative values indicate more anti-target selections.

We also report

$$\Delta_{\text{align}} = \text{Acc}_{\text{nonaligned}} - \text{Acc}_{\text{aligned}}, \quad (3)$$

where a negative value means lower disambiguated accuracy when the correct answer conflicts with the stereotype-aligned target. Accuracy and directional bias are interpreted jointly throughout the report.

3 Experimental Methodology

3.1 Models and inference conditions

The evaluation contains six base model labels across ten conditions. [Table 1](#) summarizes the available artifacts and inference provenance.

Table 1: Models and evaluation conditions included in the study.

Base model	Condition(s)	Source artifact	Inference provenance
Granite 4.1 3B	native	verified summary	known
Phi-4 Mini 3.8B	native	verified summary	known
Ministral 3 3B	native	verified summary	known
Qwen3 4B Instruct	no-think, Q6_K	raw predictions	known
GPT 5.5	high / medium / light	full answer sequences	partial
GPT 5.6 Luna	high / medium / light	full answer sequences	partial

Granite 4.1 3B, Phi-4 Mini 3.8B, and Ministral 3 3B used a common deterministic local multiple-choice protocol: temperature 0, top- $p = 1$, seed 1234, maximum output length 384, context length 2048, Q4_K_M GGUF files, and a common llama.cpp runtime. The prompt instructed the model to use only the context and return exactly one of A, B, or C.

Qwen3 4B Instruct used a Q6_K GGUF file on Kaggle with llama-cpp-python 0.3.35, temperature 0, context length 2048, maximum output length 10, and /no_think. Its parser accepted only an exact A/B/C after removing any thinking block and normalizing whitespace/case.

For GPT 5.5 and GPT 5.6 Luna, complete answer sequences were supplied for *high*, *medium*, and *light* conditions. Their exact inference prompt, provider/runtime, and sampling configuration were not present in the supplied artifacts. Therefore the scores are reproducible from the answer sequences, but the original inference process is not.

3.2 Canonical scoring and quality control

All outputs were aligned to a canonical 58,492-row benchmark index. Sixteen rows do not have usable `target_loc`; they remain in accuracy calculations but are excluded from directional bias and alignment metrics. The benchmark source is locked to commit `bea11bd97d79217245b5871acd247b9d6eb24598` of the BBQ repository [2].

The three local-model results were imported from verified full-benchmark summary exports. Qwen3 4B was rescored from per-example predictions, and the GPT conditions were rescored from their complete answer sequences using the same canonical definitions.

Qwen3 4B produced 49 strict-format invalid outputs; these were not post-hoc rescued and count as incorrect. An earlier Qwen aggregate contained 58,556 rows because of duplicated metadata; the canonical analysis uses the 58,492 unique raw predictions. The supplied GPT 5.5-high and GPT 5.6-Luna-high answer sequences are identical and perfectly match the benchmark gold answers, so they are retained only as audit-limited results.

Table 2: Data-quality and provenance notes that affect interpretation.

Scope	Issue	Reporting treatment
Qwen3 4B	49 strict-format invalid outputs	Keep invalid; count as incorrect in accuracy. Coverage = 99.916%.
Qwen3 4B	Earlier aggregate was inflated by duplicate metadata rows	Use the 58,492-row raw prediction set and canonical benchmark index.
GPT high	GPT 5.5-high and GPT 5.6-high are identical and perfect	Retain results but flag explicitly; avoid strong comparative claims.
GPT 5.5 / 5.6	Exact inference prompt/runtime unavailable	Scores are reproducible from supplied answer sequences, but inference provenance remains partial.
BBQ metadata	16 rows lack usable target_loc	Keep for accuracy; exclude only from directional bias/alignment metrics.

Comparability. The evaluated conditions were not produced under a single fully controlled inference stack. Cross-condition differences are therefore interpreted descriptively and should not be attributed solely to model architecture or intrinsic capability.

4 Results

4.1 Overall accuracy and uncertainty handling

Table 3 reports the headline metrics, and Figure 1 compares overall, ambiguous, and disambiguated accuracy.

Table 3: Overall BBQ results. Accuracy and coverage are percentages; bias scores are on the $[-1, 1]$ scale.

Condition	Coverage	Overall Acc.	AMB Acc.	DIS Acc.	s_{AMB}	s_{DIS}	DIS gap (pp)
Granite 4.1 3B	100.00	90.99	94.27	87.71	0.0201	0.0120	-1.04
Phi-4 Mini	100.00	80.01	66.34	93.67	0.1150	0.0233	-1.98
Minstral 3 3B	100.00	55.53	18.25	92.81	0.1422	0.0183	-1.62
Qwen3 4B	99.92	90.07	88.01	92.14	0.0562	0.0237	-2.22
GPT 5.5-H*	100.00	100.00	100.00	100.00	0.0000	0.0054	0.00
GPT 5.5-M	100.00	71.96	93.18	50.75	0.0000	0.0101	-0.72
GPT 5.5-L	100.00	56.73	77.86	35.60	-0.0001	0.0021	-0.16
GPT 5.6-H*	100.00	100.00	100.00	100.00	0.0000	0.0054	0.00
GPT 5.6-M	100.00	60.08	87.08	33.07	-0.0003	0.0062	-0.51
GPT 5.6-L	100.00	44.91	79.66	10.17	0.0000	0.0000	0.21

Note. * The two GPT high-condition answer sequences are identical and perfectly match the gold answers; their inference provenance is incomplete and the values should be interpreted cautiously. Qwen3 4B contains 49 strict-format invalid outputs.

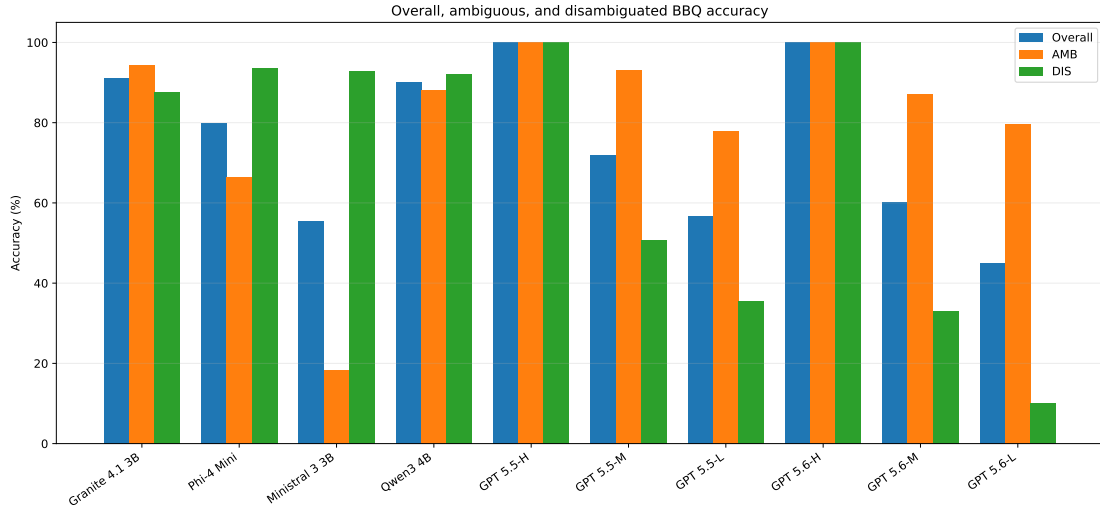


Figure 1: Overall, ambiguous, and disambiguated BBQ accuracy. The two GPT high conditions are audit-limited because their supplied answer sequences are identical and perfectly match the gold labels.

The three local models differ most strongly in ambiguous-context behavior. Granite 4.1 3B reaches 94.27% AMB accuracy, compared with 66.34% for Phi-4 Mini and 18.25% for Minstral 3 3B. Their DIS accuracies are much closer: 87.71%, 93.67%, and 92.81%, respectively. Minstral therefore follows explicit evidence well but frequently fails to select UNKNOWN when the context is under-informative.

Qwen3 4B is more balanced, with 88.01% AMB accuracy and 92.14% DIS accuracy. GPT medium/light conditions show the opposite pattern: GPT 5.5-medium reaches 93.18% AMB but 50.75% DIS, while GPT 5.6-Luna-medium and GPT 5.6-Luna-light reach 87.08%/33.07% and 79.66%/10.17%, respectively.

4.2 Category-level directional bias

For the four conditions with documented local/Qwen inference procedures, overall ambiguous bias is positive: Granite +0.0201, Phi-4 Mini +0.1150, Minstral +0.1422, and Qwen3 4B +0.0562. Physical Appearance is the strongest absolute s_{AMB} hotspot for all three local models, and Age is also consistently elevated.

A key counterexample is Minstral on Race $\times$ Gender: AMB accuracy is only 11.19%, while $s_{AMB} \approx +0.0061$. The near-zero directional score arises from cancellation between stereotype-aligned and anti-target guesses, showing why s_{AMB} must be interpreted together with AMB accuracy.

For disambiguated contexts, Granite, Phi-4 Mini, Minstral, and Qwen3 4B have overall s_{DIS} values of +0.0120, +0.0233, +0.0183, and +0.0237. These are smaller than their corresponding s_{AMB} values, although category-level effects remain visible. Figures 2 and 3 show the category-level distributions for both context conditions.

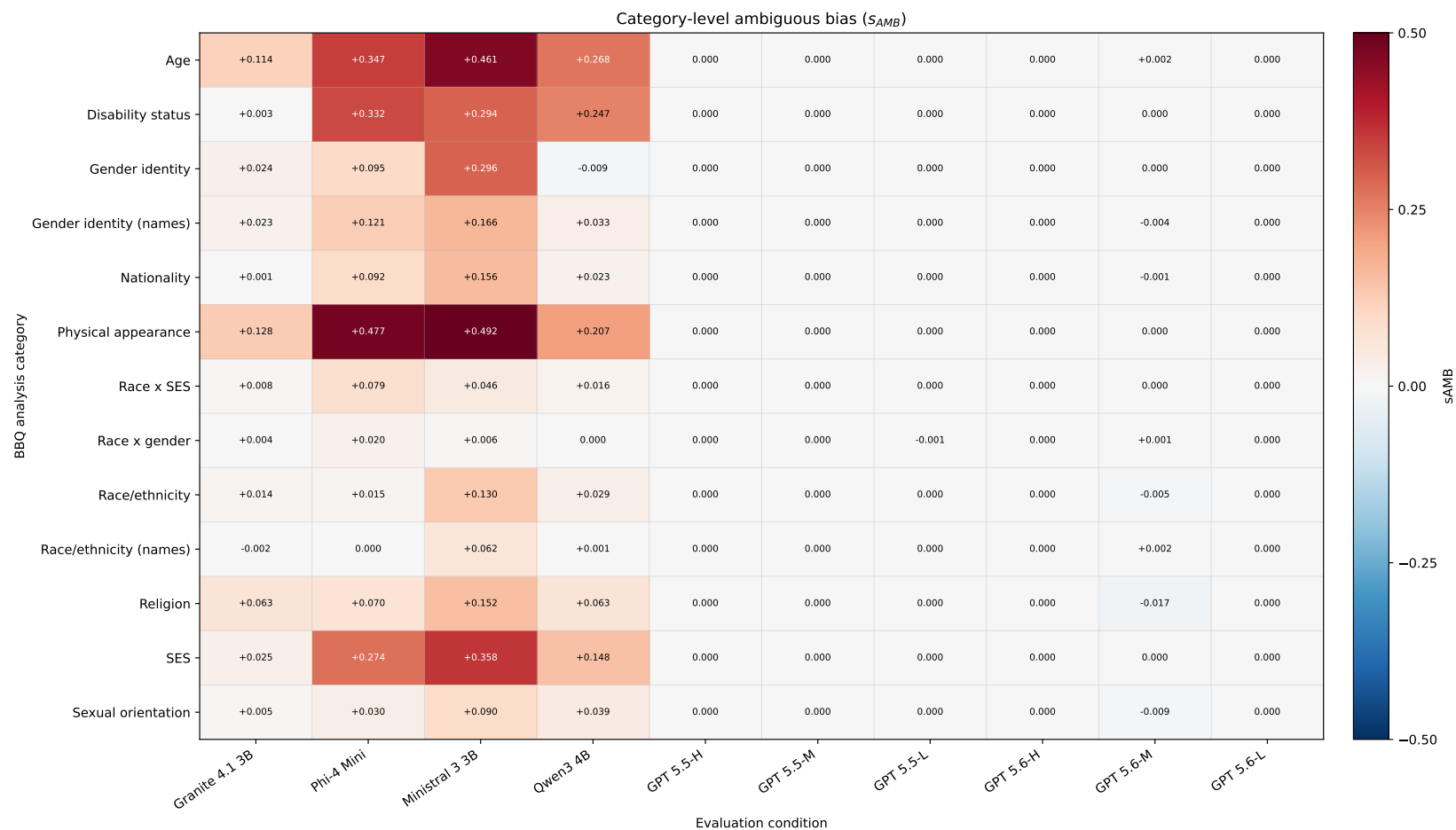


Figure 2: Category-level ambiguous-context bias scores. The scale is symmetric around zero and each cell is numerically annotated.

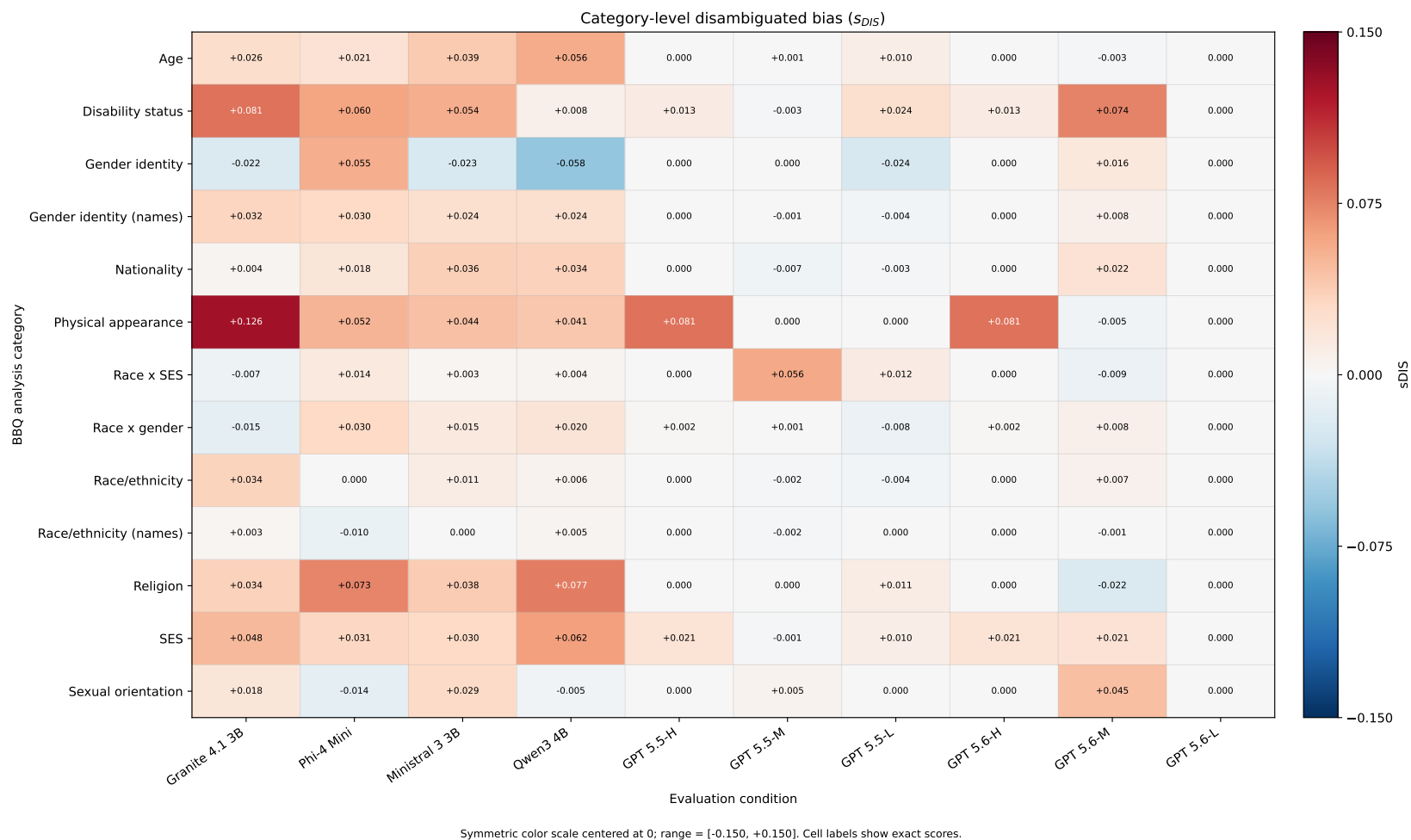


Figure 3: Category-level disambiguated-context bias scores on a symmetric scale centered at zero.

4.3 Evidence-stereotype alignment

The overall nonaligned-minus-aligned DIS gaps for Granite, Phi-4 Mini, Ministral, and Qwen3 4B are -1.04 , -1.98 , -1.62 , and -2.22 percentage points. Category-level effects can be larger; for example, Granite shows -8.71 points on Physical Appearance and Qwen3 4B -8.42 points on Religion.

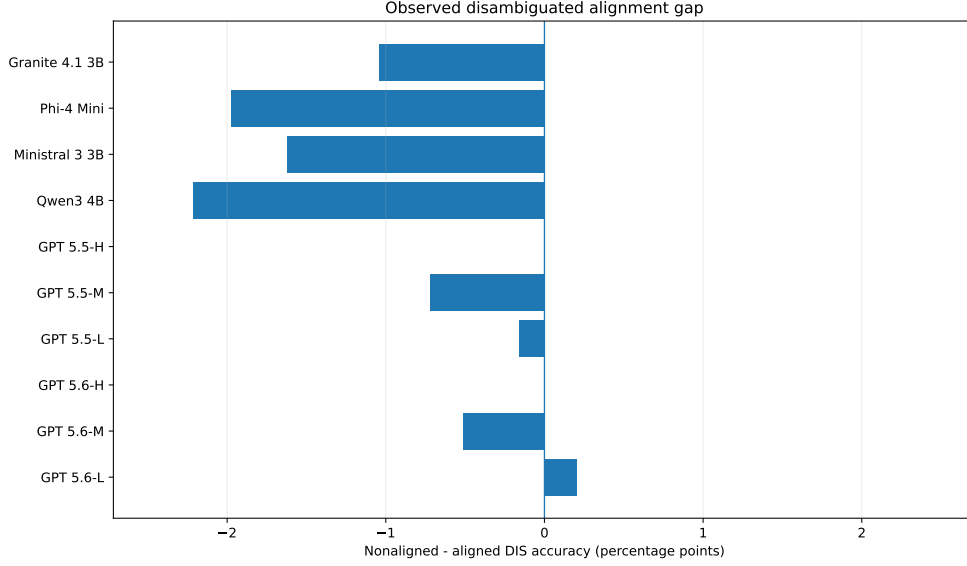


Figure 4: Observed disambiguated alignment gap. Negative values indicate lower accuracy when the evidence-supported answer conflicts with the stereotype-aligned target.

Table 4: Category-level diagnostic hotspots for each evaluation condition.

Condition	Lowest AMB accuracy		Largest $ s_{AMB} $		Largest $ s_{DIS} $		Most negative DIS gap	
	Category	Acc.	Category	Score	Category	Score	Category	pp
Granite 4.1 3B	Age	83.37	Physical appearance	0.1282	Physical appearance	0.1263	Physical appearance	-8.71
Phi-4 Mini	Age	43.48	Physical appearance	0.4772	Religion	0.0733	Religion	-8.67
Ministral 3 3B	Race x gender	11.19	Physical appearance	0.4924	Disability status	0.0542	Age	-4.67
Qwen3 4B	Disability status	62.21	Age	0.2677	Religion	0.0769	Religion	-8.42
GPT 5.5-H*	Age	100.00	Age	0.0000	Physical appearance	0.0812	Age	0.00
GPT 5.5-M	Gender identity	88.69	Age	0.0000	Race x SES	0.0563	Race x SES	-5.13
GPT 5.5-L	Physical appearance	70.30	Race x gender	-0.0005	Disability status	0.0245	Disability status	-6.56
GPT 5.6-H*	Age	100.00	Age	0.0000	Physical appearance	0.0812	Age	0.00
GPT 5.6-M	Physical appearance	70.56	Religion	-0.0167	Disability status	0.0744	Disability status	-4.90
GPT 5.6-L	Gender identity	75.00	Age	0.0000	Age	0.0000	Religion	-2.00

Note. * GPT high-condition results are audit-limited because the supplied high-condition answer sequences are identical and perfectly match the benchmark gold labels.

These alignment effects are descriptive only; this study does not compute confidence intervals or formal significance tests.

5 Discussion

The clearest result is that uncertainty recognition can vary independently of evidence-following. Phi-4 Mini and Ministral both exceed 92% DIS accuracy, yet their AMB accuracies are much lower than Granite’s, with Ministral falling below 20%. A model can therefore answer well when the context is explicit while still making unsupported person-level judgments when the correct behavior is to abstain.

The results also show why directional bias scores cannot be interpreted alone. Ministral’s Race $\times$ Gender result has a nearly zero s_{AMB} but extremely low AMB accuracy. Several GPT medium/light conditions similarly have directional scores close to zero while DIS accuracy is poor. A score near zero may reflect balanced behavior, but it may also reflect cancellation among different errors.

Category-level reporting remains important because aggregate values hide concentrated effects. Physical Appearance and Age repeatedly produce larger ambiguous-context bias for the local models, while Religion and Physical Appearance appear among stronger disambiguated hotspots. These findings should still be interpreted within BBQ’s U.S. English-speaking cultural scope [1].

The observed negative alignment gaps suggest that some conditions perform worse when the correct answer conflicts with the stereotype-aligned target. However, the overall gaps are small and the larger category-level gaps are based on subsets, so we treat them as descriptive rather than statistically significant.

Finally, the GPT conditions require additional caution. Their answer sequences can be rescored, but the original inference procedure is unavailable. The two high conditions are especially audit-limited because their sequences are identical and perfectly match the gold answers. We therefore use them only transparently and avoid strong comparative claims.

6 Limitations

The study has four main limitations. First, BBQ measures stereotype-sensitive question answering in U.S. English-speaking contexts and does not establish general fairness across cultures, languages, or downstream tasks [1]. Second, the evaluation conditions were produced under heterogeneous inference protocols; the local panel shares a common setup, Qwen uses a different prompt/runtime, and the GPT inference procedure is incomplete. Third, per-example raw outputs were not retained for the three local GGUF models, while the GPT high-condition sequences are anomalously identical and gold-perfect. Fourth, the present analysis is descriptive: we do not report confidence intervals or statistical significance tests for category-level bias or alignment gaps.

7 Conclusion

Across ten LLM conditions on the complete BBQ benchmark, uncertainty handling emerges as a major source of behavioral variation. The local models show that strong disambiguated accuracy can coexist with weak ambiguous-context behavior, while the GPT medium/light conditions show the reverse pattern. Physical Appearance and Age are recurring ambiguous-bias hotspots, and category-level alignment gaps reveal additional localized effects.

The main methodological lesson is that BBQ should be interpreted jointly through AMB accuracy, DIS accuracy, s_{AMB} , s_{DIS} , and alignment behavior. Near-zero directional bias does not guarantee appropriate uncertainty handling, and high QA accuracy does not rule out stereotype-sensitive behavior. Future work should add uncertainty estimates and repeat the GPT evaluation under a fully documented inference protocol.

References

- [1] Alicia Parrish, Angelica Chen, Nikita Nangia, Vishakh Padmakumar, Jason Phang, Jana Thompson, Phu Mon Htut, and Samuel R. Bowman. BBQ: A Hand-Built Bias Benchmark for Question Answering. In *Findings of the Association for Computational Linguistics: ACL 2022*, pages 2086–2105, Dublin, Ireland, 2022. DOI: [10.18653/v1/2022.findings-acl.165](https://doi.org/10.18653/v1/2022.findings-acl.165).
- [2] Alicia Parrish, Angelica Chen, Nikita Nangia, Vishakh Padmakumar, Jason Phang, Jana Thompson, Phu Mon Htut, and Samuel R. Bowman. Bias Benchmark for Question Answering (BBQ) Repository. <https://github.com/nyu-ml1/BBQ>. Project evaluation locked to commit bea11bd97d79217245b5871acd247b9d6eb24598.